

Job Board API & Dashboard

Role: Software Developer

Difficulty: junior

Project Type: fullstack

Duration: 14 days

Focus: api_development

Candidate: Rahul Sharma

Project Overview

Build a RESTful job posting API with JWT auth and a Next.js dashboard that lists, creates, updates, deletes jobs and exports job details to PDF.

Technologies

- JavaScript
- TypeScript
- React
- Next.js
- Node.js
- Express.js
- Tailwind CSS
- MongoDB

Requirements

- Create secure JWT auth
- Implement CRUD for jobs
- Export job to PDF
- Use Next.js for UI
- Document API

Objectives

- Design API schema
- Implement auth
- CRUD jobs
- Build dashboard
- Export PDF

Functional Requirements

- User register/login
- Role-based access
- Create/read/update/delete jobs
- Paginated job list
- Search by title
- PDF export

Non-Functional Requirements

- JWT security
- Input validation
- Error handling
- Responsive UI
- Swagger docs

Modules

Auth

Register, login, token refresh

- register
- login
- refresh

Job

CRUD and search

- create
- read
- update
- delete
- search

PDF

Generate PDF of job

- generate
- download

Frontend

Next.js UI

- pages
- components

API Client

Axios wrapper

- get
- post
- put
- delete

Utils

Validation & error handling

- validate
- handleError

Implementation Plan

Setup

- init repo
- install deps
- setup CI

Schema

- design MongoDB models
- create API routes

Auth

- implement register/login
- JWT middleware

Job

- CRUD endpoints
- search logic

PDF

- PDF generation
- download route

Frontend

- build pages
- consume API
- test

Evaluation Criteria

API design (20%)

RESTful, clear endpoints

Security (20%)

JWT, role checks

Frontend usability (15%)

Responsive, intuitive UI

Code quality (15%)

Clean, documented

Testing (15%)

Unit & integration tests

Performance (15%)

Scalable, efficient

Deliverables

- GitHub repo
- Postman collection
- Swagger docs
- PDF sample
- README
- Demo video

Suggested Folder Structure

- /backend
- /frontend
- /docs
- /scripts
- /tests
- /config

API Endpoints

POST /api/auth/register

user registration

POST /api/auth/login

user login

GET /api/auth/me

current user

GET /api/jobs

list jobs

POST /api/jobs

create job

GET /api/jobs/:id

job details

PUT /api/jobs/:id

update job

DELETE /api/jobs/:id

delete job

GET /api/jobs/:id/pdf

export PDF

Database Design

users

store user data

- _id
- email
- passwordHash
- role
- createdAt

jobs

job postings

- _id
- title
- description
- company
- location
- postedBy
- createdAt
- updatedAt

Testing Plan

- unit tests for auth
- unit tests for job
- integration tests for API
- end-to-end tests with Cypress
- load test with k6
- security test with OWASP ZAP

Candidate Fit

Rahul's JavaScript, TypeScript, React, Next.js, Node.js, Express.js, Tailwind CSS, MongoDB, JWT, and role based auth skills align with the project, while PDF export and API documentation provide new learning opportunities.